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AIM:Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for
- a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree

df = pd.read_csv("C:/Users/hrane/Downloads/diabetes_risk_prediction_073.csv")

print("Dataset Loaded successfully. Shape:", df.shape)

df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

df = df.dropna()

print("Target classes found:", df["Diabetes_Risk"].unique())
df["Diabetes_Risk"] = df["Diabetes_Risk"].map(
    {"Low": 0, "Moderate": 1, "High": 2}
)

columns_to_drop = [
    "Patient_ID",
    "Diabetes_Risk",
    "AI_Health_Recommendation",
    "Doctor_Consultation_Needed",
]
X_raw = df.drop(columns=columns_to_drop, errors="ignore")
y = df["Diabetes_Risk"]

X = pd.get_dummies(X_raw, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
```

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```
)

model = DecisionTreeClassifier(
    criterion="entropy", max_depth=3, random_state=42
)
model.fit(X_train, y_train)
print("\nDecision Tree trained successfully!")

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"\nAccuracy Percentage: {accuracy * 100:.2f}%")

print("\nClassification Report:")
print(
    classification_report(
        y_test, y_pred, target_names=["Low", "Moderate", "High"]
    )
)

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=list(X.columns),
    class_names=["Low", "Moderate", "High"],
    filled=True,
    rounded=True,
    fontsize=12,
)

plt.title("Decision Tree Structure - Diabetes Risk Classification", fontsize=16)
plt.tight_layout()
plt.show()
```

OUTPUT:

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```
Dataset Loaded successfully. Shape: (50000, 41)
Target classes found: <StringArray>
['Moderate', 'High', 'Low']
Length: 3, dtype: str

Decision Tree trained successfully!

Accuracy Percentage: 100.00%

Classification Report:
      precision    recall  f1-score   support

     Low       1.00      1.00      1.00         62
  Moderate       1.00      1.00      1.00        1689
     High       1.00      1.00      1.00       4785

 accuracy       1.00      1.00      1.00       6536
   macro avg       1.00      1.00      1.00       6536
 weighted avg       1.00      1.00      1.00       6536

Confusion Matrix:
[[ 62  0  0]
 [ 0 1689  0]
 [ 0  0 4785]]
```

